

PROBLEM STATEMENT: Cessna 172 Flight Simulation Setup

OBJECTIVE: Create a complete flight simulation setup for a Cessna 172 Skyhawk performing steady cruise flight

- i) 1500m altitude and Mach 0.15.
- ii) 1000m altitude and Mach 0.12

DELIVERABLES:

1. Complete aircraft configuration script
2. Custom DATCOM parser function
3. Trimmed cruise condition
4. Simulation-ready workspace

AIRCRAFT SPECIFICATIONS - CESSNA 172 SKYHAWK

MASS PROPERTIES

Empty mass: 767 kg, Maximum fuel capacity: 144 kg (53 US gallons usable), Current fuel load: 90 kg (typical cruise fuel, ~60% full), Fuel density: 720 kg/m³ (Aviation Gasoline 100LL, 6.0 lb/gal)

Moments of Inertia (body axes, about CG):

- I_{xx} (roll): 1285 kg·m²
- I_{yy} (pitch): 1824 kg·m²
- I_{zz} (yaw): 2666 kg·m²
- I_{xz} (cross-product): 0 kg·m²
- I_{xy} : 0 kg·m²
- I_{yz} : 0 kg·m²

Center of Gravity Location (body frame):

- x_{cg} : 2.11 m (aft of nose)
- y_{cg} : 0.00 m (centerline)
- z_{cg} : 1.26 m (above fuselage reference)

WING GEOMETRY

Wing Configuration: High-wing, rectangular-tapered Wing Span (b): 10.98 m , Wing Area (S): 16.17 m² (174 sq ft), Mean Aerodynamic Chord ($\bar{c}$): 1.50 m, Root Chord (c_r): 1.626 m , Tip Chord (c_t): 1.13 m , Taper Ratio (λ): 0.695

Reference Dimensions:

- S_{ref} = wing_area
- b_{ref} = wing_span
- c_{ref} = mean_aerodynamic_chord

AERODYNAMIC DATA

Data Source: USAF DATCOM output file File Location: "cessna.out"

Required Parser Function: c172datcom(filepath) Input: DATCOM output file path Output: Lookup function handle

Parser Requirements:

1. Extract static coefficients (CL, CD, CM) vs alpha
2. Extract stability derivatives (CLA, CMA, CYB, CNB, CLB)
3. Extract dynamic derivatives (CLQ, CMQ, CLP, CNR, CLR, CYP, CNP)

Default Values for Missing Data:

- CLA = 4.7
- CMA = -0.9
- CYB = -0.31
- CNB = 0.071
- CLB = -0.089
- CLQ = 3.9
- CMQ = -12.4
- CLP = -0.49
- CNR = -0.095
- CLR = 0.14
- CYP = -0.19
- CNP = -0.03

Control Surface Effectiveness:

- $CLDE = 0.30$
- $CMDE = -1.20$
- $CLDA = 0.075$
- $CNDA = 0.008$
- $CYDR = 0.20$
- $CNDR = -0.075$

CONTROL SURFACES

Ailerons:

- Type: Primary roll control
- Axis: $[1 \ 0 \ 0]$
- Maximum deflection: $+20^\circ$ (0.349 rad)
- Minimum deflection: -20° (-0.349 rad)
- Roll moment derivative: 0
- Pitch moment effect: 0
- Yaw moment effect: 0

Elevator:

- Type: Primary pitch control
- Axis: $[0 \ 1 \ 0]$
- Maximum deflection: $+28^\circ$ (0.489 rad)
- Minimum deflection: -26° (-0.454 rad)
- Lift effect: 0.30
- Pitch moment: -1.20
- Yaw moment effect: 0

Rudder:

- Type: Primary yaw control
- Axis: $[0 \ 0 \ 1]$

- Maximum deflection: $+30^\circ$ (0.524 rad)
- Minimum deflection: -30° (-0.524 rad)
- Roll moment effect: 0
- Pitch moment effect: 0
- Yaw moment: -0.075

PROPULSION SYSTEM

Engine Model: Lycoming O-360-A1A2 Engine Type: 4-cylinder, horizontally-opposed, naturally aspirated Rated Power: 134,228 W (180 BHP @ 2700 RPM) Displacement: 361 cubic inches (5.916 liters) Bore $\times$ Stroke: 5.125 in $\times$ 4.375 in (130.2 mm $\times$ 111.1 mm) Compression Ratio: 8.5:1 , Max Continuous RPM: 2700 RPM , Dry Weight: 131 kg (288-291 lb)

Propeller Configuration:

- Type: Fixed-pitch, two-blade
- Element Type: 'propeller'
- Diameter (D): 1.905 m (75 inches)
- Pitch: 1.219 m (48 inches)
- Number of Blades: 2
- Efficiency: 0.80
- Thrust Coefficient (C_t): [0.10, -0.05]
- Power Coefficient (C_p): [0.05, 0.02]
- Disk Loading Factor: 0.05
- Induced Velocity Factor: 0.1

Engine Location (body frame):

- x: 1.68 m (forward of CG)
- y: 0.00 m (centerline)
- z: 1.26 m (at CG height)

Thrust Direction: [1 0 0] Fuel Flow Rate: 0 (computed by propeller model) Thrust Model Function: [] (empty)

FLIGHT CONDITION

Flight Phase: Steady level cruise Altitude (h): 1000 m MSL Mach Number (M): 0.15 True
Airspeed (TAS): Calculate from $V = M \times a(h)$ Flight Path Angle (γ): 0° (level flight) Simulation
Duration: 120 seconds

TRIM REQUIREMENTS

Trim Solver Configuration:

- Tolerance: 1×10^{-3}
- Maximum Iterations: 15,000
- Optimizer: fmincon (use_fmincon = true)
- Initial Guess:
 - Alpha: 2.0° (0.0349 rad)
 - Elevator: -1.0° (-0.0175 rad)
 - Throttle: 0.65 (65%)

Trim Objective:

- Sum of forces: [0, 0, 0] N (± 1 N acceptable)
- Sum of moments: [0, 0, 0] N·m (± 0.01 N·m acceptable)

SIMULATION PARAMETERS

Simulink Model: 'Flightsim_aircraft' , Fixed Time Step: 0.01 s , Simulation Time: 120 s, Sample
Points: 1000

REQUIRED WORKSPACE VARIABLES

Must assign to base workspace:

- ac (Aircraft object)
- initial_state (12×1 trim state vector)
- Initialpos (Initial position [x, y, z])
- InitialVel (Initial velocity [u, v, w])
- InitialOri (Initial orientation [ϕ , θ , ψ])
- InitialRot (Initial rates [p, q, r])
- n_cs (Number of control surfaces: 3)
- n_pe (Number of propulsive elements: 1)

- `n_total` (Total control inputs: 4)
- `control_input_data` ([time, controls] array)
- `sim_stop_time` (Simulation duration)

TASKS

1. Create `c172datcom.m` parser function
2. Parse the DATCOM file and extract aerodynamic data
3. Configure the complete Cessna 172 aircraft
4. Set up the propeller propulsion system
5. Solve for trim condition at 1500m, Mach 0.15 and 1000m, Mach 0.12
6. Verify trim solution
7. Generate control schedule for 120-second cruise
8. Assign all required variables to workspace
9. Document trim results

EVALUATION CRITERIA

- Parser correctly extracts all aerodynamic coefficients
- Aircraft configuration matches specifications exactly
- Propeller configured with correct parameters
- Trim solution converges (`converged = true`)
- All workspace variables correctly assigned
- Code is clean and organized
- Simulation runs without errors for full 120 seconds